

A2-01 Wall-Wart Transient Capture — DS1054Z Procedure

A2-01 Wall-Wart Transient Capture

Bench-side reference for capturing the plug-in transient on +VRAW using the **Rigol DS1054Z** oscilloscope. Purpose: validate the hypothesis that the 24 VAC wall wart's plug-in overshoot exceeds the LM2596's 45 V absolute max, killing the chip on first power-up.

Test board: **A2-01** (LM2596 already dead — nothing to lose from repeated plug-in events, and the failure conditions are directly reproducible).

Companion memory: `project_a2_01_lm2596_failure.md`. Companion log: `poc/rev2/dashboard.md`.

Equipment

- ☐ Rigol DS1054Z scope, powered on
 - ☐ 1× 10× passive probe (comes with the scope)
 - ☐ Phone with camera (USB writes are blocked by CrowdStrike DLP — using photos instead of scope-saved PNGs)
 - ☐ Switched outlet strip (rocker switch), for repeatable plug-in events — recommended
 - ☐ 24 VAC wall wart (the one that killed A2-01)
 - ☐ Board A2-01 with barrel jack accessible and R_JMP2 pad 1 reachable
 - ☐ Small paper label + pen (write “T01”, “T02”, ... to hold in-frame per sample)
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Board prep

- ☐ Wall wart **unplugged from wall AND from board**
 - ☐ R_JMP1 installed (or wire-jumped) — completes 24 VAC → bridge → C5 path
 - ☐ R_JMP2, R_JMP3, R_JMP4 all **out** — isolates the (dead) LM2596 from the transient we're characterizing
 - ☐ Nothing else connected to the board (no USB pigtail, no zone loads)
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Probe preparation (one-time)

Compensate the probe against the scope's built-in reference square wave:

1. Confirm probe body switch is set to **10×** (not 1×)
 2. Plug probe BNC into **CH1** input
 3. Clip probe tip to the front-panel **Probe Comp** post (below CH1); ground clip to the adjacent **GND** post
 4. Press **AUTO** (top row). Scope displays the 1 kHz square wave
 5. Adjust the trimmer screw on the probe body until corners are square — no overshoot, no rounding
 6. Press **CH1** button → **Probe** softkey → select **10×**
 - This tells the scope to display real-world voltages, not the 1/10 attenuation at the BNC
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Channel setup (CH1)

Press **CH1** to open the channel menu. Softkeys on the right of the display:

Softkey	Set to
Coupling	DC
BW Limit	OFF
Probe	10x
Invert	OFF
Vernier	OFF
Unit	Voltage

Vertical scale: turn the outer **SCALE** knob under CH1 until display reads **10.0 V/div** (see indicator at bottom-left).

Vertical position: turn the inner **POSITION** knob to move the zero-line to ~1 division from the bottom. Steady-state 30 V will land near center, leaving 5 divisions of headroom for spikes.

Horizontal (timebase) setup

Big **SCALE** knob in the HORIZONTAL cluster (center of front panel):

- ☐ Set to **10 ms/div** for initial capture (120 ms full-screen window, ~7 AC cycles at 60 Hz)

Horizontal **POSITION** knob: shift the trigger indicator (**T** marker at top of screen) to sit ~10 % from the left edge, leaving 90 % post-trigger.

Memory depth + acquisition

Press **Acquire** (top-right button cluster):

Softkey	Set to
Acquisition	Normal (never Averaging — kills single-shot transients)
Mem Depth	12 M (deepest UltraVision setting; keeps sample rate high at slow timebase)
AntiAlias	OFF

At 10 ms/div with 12 Mpts, sample rate stays ~100 MSa/s → resolves any μ s-scale transient cleanly.

Trigger setup

Press **MENU** button in the TRIGGER cluster (right of front panel, below trigger LEVEL knob):

Softkey	Set to
Type	Edge
Source	CH1
Slope	↑ Rising
Coupling	DC
Noise Reject	ON
HF Reject	OFF
Holdoff	(default, 16 ns is fine)

Trigger level: turn the dedicated **LEVEL** knob (small knob in trigger cluster) until the top-right of the screen reads **+5.0 V**. The dashed trigger line on the right of the plot moves as you turn.

Trigger mode: press **MODE** button (upper right of trigger cluster) until the Auto / Normal / Single LED strip lights **Normal** (middle LED).

Probe placement

Target net: **+VRAW**. Any pad on that net reads the same voltage during the transient. The physically-accessible option is **R_JMP2 pad 1** — the +VRAW-side pad of R_JMP2. Since R_JMP2 stays out for this test, R_JMP2 pad 1 is a bare copper pad, easy to grip with the probe hook.

Identify pad 1 vs pad 2 (silkscreen isn't always clear):

- ☐ DMM in continuity/beep mode
- ☐ One probe on C5(+) top
- ☐ Touch each R_JMP2 pad in turn — the one that **beeps (0 Ω)** is pad 1 (=+VRAW). The other (pad 2) reads open (floats — do NOT probe there)
- ☐ Optional: mark pad 1 with a Sharpie dot on the silkscreen

Then:

- ☐ Probe tip → **R_JMP2 pad 1** (verified via continuity to C5+)
- ☐ Ground clip → any solid GND on the board:
 - Mounting hole ring (bottom-left, exposed copper), OR
 - D3 anode terminal, OR
 - C5(−) cap lead
- ☐ Alligator clip, not the sprung tip — better ground return, less noise on fast edges

With no power applied, scope should show a flat trace at 0 V (or very close). If it drifts, press **CH1** twice quickly to auto-zero.

Do NOT install R_JMP2 for this test. A2-01's dead LM2596 (~9 Ω effective short from VIN to GND under power) would clamp +VRAW to near-ground on plug-in, masking the transient. Keep R_JMP2 out so C5 alone loads the rail and the overshoot rings out cleanly for the scope to catch.

Single-shot capture procedure

1. Board bench-mounted, wall wart's DC output plug in the barrel jack, wall wart's AC plug **not yet in outlet** (or outlet strip switch OFF)
2. Verify scope shows flat 0 V on CH1

3. Press **SINGLE** (top-right button, next to RUN/STOP)
 - Top-of-screen status changes to **WAIT** / armed
 - RUN/STOP LED goes red
 4. **Plug the wall wart into the outlet** (or flip the outlet strip switch)
 5. Scope triggers, captures, freezes
 - Status changes to **STOP**; the transient is now held in memory
 6. If it triggered on noise instead of the real event:
 - Press **CLEAR**
 - Press **SINGLE** to rearm
 - Unplug and replug (or cycle the outlet strip switch)
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Read the peak

Quick way — auto-measure:

1. Press **Measure** button
2. Softkey **Voltage** → **Vmax**
3. Reading appears at bottom of screen

Precise way — manual cursors:

1. Press **Cursor** button
 2. **Mode** → **Manual**
 3. **Type** → **Y** (voltage cursors)
 4. Turn the multi-purpose knob (top-left, labeled **Intensity** when idle) to move **CurA** to the peak of the transient
 5. Softkey to switch cursor selection to **CurB**, move to 0 V baseline
 6. Read **BY-AY** at the bottom — peak voltage
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Multi-sample procedure

The transient magnitude depends on the AC line phase at plug-in — worst case is plug-in at line peak. Capture 10–20 samples:

1. Note the peak value on paper (see log grid below)
2. Press **CLEAR**
3. Press **SINGLE** to rearm
4. Cycle the outlet strip switch OFF, then ON
5. Repeat

Best practice: save the **worst-case** capture as an image (see next section).

Saving captures — phone photo

USB mass-storage writes are blocked by CrowdStrike DLP on the workstation, so we're photographing the scope screen instead of writing PNGs to a USB stick. It's ugly but it captures what we need: the trace shape, the Vmax measurement box, and a paper label for sample ID.

Per sample:

1. Write the sample number on a scrap of paper: **T01**, **T02**, **T03**, ...

2. Hold the label just below the scope screen (visible in frame, doesn't obscure the trace)
3. Photograph **square-on**, not at an angle:
 - Stand directly in front of the scope
 - Fill the frame with the display bezel-to-bezel
 - Keep the phone parallel to the scope face — avoids keystone distortion
4. Verify in the shot:
 - Trace is visible
 - `Vmax` value at the bottom is legible
 - `10.0V/div` scale indicator at bottom-left is legible
 - Paper sample label is readable

Lighting tips:

- The DS1054Z's TFT screen glares under overhead LEDs. If you see hotspots in the preview, tilt the phone slightly or shade the screen with a piece of cardboard
- Avoid using flash — it washes out the display and creates a hard reflection
- Higher-contrast phone modes (some cameras have a “documents” mode) help legibility

Batch transfer to Mac:

1. Collect all photos in a single session (do the whole 15-sample run before transferring)
2. Select all shots in the phone's Photos app
3. AirDrop to Mac → they land in `~/Downloads/`
4. Rename in bulk, using the naming convention below

Filename convention:

```
A2-01_T<sample>_<Vmax>V.jpg
```

Examples:

```
A2-01_T01_38V.jpg
A2-01_T02_45V.jpg
A2-01_T03_51V.jpg
```

Landing location:

```
poc/rev2/board-A2-01-transient-capture/
├─ A2-01_T01_38V.jpg
├─ A2-01_T02_45V.jpg
└─ ...
```

Reference the worst-case shot in `poc/rev2/dashboard.md` under A2-01 Anomalies with the filename and `Vmax`.

Interpretation

Vmax	Diagnosis
32–35 V	Plug-in event alone did NOT kill the chip. Theory partially wrong — look for cumulative stress, marginal silicon, or another mechanism.
36–44 V	Marginal — right at abs max but not clearly over. Plausible but not decisive.
45–60 V	✅ Theory validated. TVS installation is fully justified. SMBJ33CA-13-F (53 V max clamp) is well matched.
>60 V	Theory validated AND SMBJ33CA's clamping voltage is marginal. Reconsider unidirectional SMBJ30A or lower Vrwm part for rev3.

Log the worst-case Vmax and the count of samples in the field-log below.

Field log

Fill in on paper as you go. Copy back to `dashboard.md` under A2-01 Anomalies afterward.

Date: ____-____-____ Time: ____:____ Ambient: ____ °F

Wall wart AC no-load: _____ VAC (DMM)

Sample #	Vmax (V)	Notes
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1	_____	_____
2	_____	_____
3	_____	_____
4	_____	_____
5	_____	_____
6	_____	_____
7	_____	_____
8	_____	_____
9	_____	_____
10	_____	_____
11	_____	_____
12	_____	_____
13	_____	_____
14	_____	_____
15	_____	_____

Worst-case Vmax: _____ V Photo filename: _____

Interpretation: ☐ 32–35 V ☐ 36–44 V
 ☐ 45–60 V ☒ ☐ >60 V

Verdict: _____

Safety

- DS1054Z front-panel input rating: 1 M Ω // 13 pF, 300 V RMS CAT I. With a 10 $\times$ probe (300 V CAT II), you have >600 V effective headroom over the ~40 V under test. Zero risk of scope input damage.
 - Hands off bare board pads while wall wart is plugged into the outlet. The 24 VAC secondary (~34 V peak) can shock through a scratch or wet skin.
 - Never touch the scope's ground clip to any un-isolated AC-mains reference. In this test, ground is the board's GND — the secondary side of the transformer — which is floating and safe.
 - If the scope's trigger LED stops flashing but the trace is off-screen, adjust vertical scale or position — don't assume no signal.
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Notes for future revs

If Vmax comes in above 45 V: - Confirms rev3 needs LM2596HV (57 V abs max) OR a proper SMB TVS footprint in place — see `project_rev3_design_notes.md` - If Vmax > 60 V, upgrade to unidirectional SMBJ30A or add an SMBJ36CA in parallel

If Vmax stays below 40 V across 20 samples: - The A2-01 chip may have been a clone-silicon defect rather than a transient event - Consider deeper root-cause analysis before finalizing rev3 TVS placement

Either way, keep the raw PNG captures — they're the empirical basis for the rev3 buck-input design.